that had never before been seen in Earth labs. Nebulium was simply the signature of ordinary oxygen doing extraordinary things.

This universality of physical laws tells us that if we land on another planet with a thriving alien civilization, they will be running on the same laws that we have discovered and tested here on Earth—even if the aliens harbor different social and political beliefs. Furthermore, if you wanted to talk to the aliens, you can bet they don't speak English or French or even Mandarin. Nor would you know whether shaking their hands—if indeed their outstretched appendage is a hand—would be considered an act of war or of peace. Your best hope is to find a way to communicate using the language of science.

Such an attempt was made in the 1970s with *Pioneer 10* and *11* and *Voyager 1* and *2*. All four spacecraft were endowed with enough energy, after gravity assists from the giant planets, to escape the solar system entirely.

Pioneer wore a golden etched plaque that showed, in scientific pictograms, the layout of our solar system, our location in the Milky Way galaxy, and the structure of the hydrogen atom. *Voyager* went further and also included a gold record album containing diverse sounds from mother Earth, including the human heartbeat, whale "songs," and musical selections from around the world, including the works of Beethoven and Chuck Berry. While this humanized the message, it's not clear whether alien ears would have a clue what they were listening to—assuming they have ears in the first place. My favorite parody of this gesture was a skit on NBC's *Saturday Night Live*, shortly after the *Voyager* launch, in which they showed a written reply from the aliens who recovered the spacecraft. The note simply requested, "Send more Chuck Berry."

Science thrives not only on the universality of physical laws but also on the existence and persistence of physical constants. The constant of gravitation, known by most scientists as "big G," supplies Newton's equation of gravity with the measure of how strong the force will be. This quantity has been implicitly tested for variation over eons. If you do the math, you can determine that a star's luminosity is steeply dependent on big G. In other words, if big G had been even slightly different in the past, then the energy output of the Sun would have been far more variable than anything the biological, climatological, or geological records indicate.

Such is the uniformity of our universe.